



$$-\frac{d[\text{Tl}^{3+}]}{dt} = \frac{k [\text{Fe}^{2+}]^2 [\text{Tl}^{3+}]}{[\text{Fe}^{2+}] + k' [\text{Fe}^{3+}]}$$

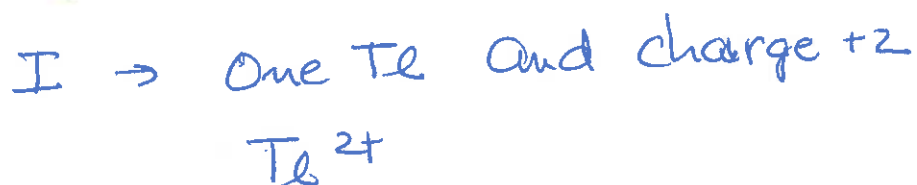


Assume 2nd step is slow

$$r = \frac{k [\text{Fe}^{2+}]^2 [\text{Tl}^{3+}]}{[\text{Fe}^{3+}]}$$



$$\text{Charge of RDS} = 4 + 3 - 3 = 4$$





$$r = \frac{k [\text{C}(\text{CH}_3)_3\text{Cl}] [\text{OH}^-]}{k' [\text{Cl}^-] + k'' [\text{OH}^-]}$$



$$k'' \ll k'$$

$$r = \frac{k [\text{C}(\text{CH}_3)_3\text{Cl}] [\text{OH}^-]}{k' [\text{Cl}^-]}$$



Charge $\rightarrow 0$





$$\text{Rate} = k \frac{[\text{Hg}_2^{2+}][\text{Te}^{3+}]}{[\text{Hg}^{2+}]}$$



Stoichiometry



Change $2+3-2 = 3$





$$\frac{d(\text{Ph}_2\text{CHOH})}{dt} = \frac{\alpha [\text{Ph}_2\text{CHCl}]}{\beta + [\text{Cl}^-]}$$



Stoichiometry of RDS = Ph_2CH

$$\text{Charge} = +1$$



④ Unimolecular Reaction

$$\text{Rate law} = \frac{k[A]^2}{k'[A] + k''}$$

High Concentration of $[A]$

First order reaction

Low Concentration of $[A]$

Second order reaction



$$k'' \ll k'$$

Stoichiometry $\rightarrow$ one A

$$k' \ll k''$$

Stoichiometry $\rightarrow$ two A



Stoichiometry of I is A